

THE FOUR-PHASE AUTHORITY HANDSHAKE PROTOCOL & DISCRETE STATE ARBITER

Deterministic 1 kHz Nervous-System Handshake: Autonomous Propose → Hardware Intercept → Operator Acknowledge → Bumpless Ramp → Active Authority

FOUR-PHASE AUTHORITY HANDSHAKE PIPELINE (1 kHz SYNCHRONOUS BUS)

PHASE 1 · REQUEST

Initiation Trigger

- Operator torque $|\tau| > 4.0 \text{ N}\cdot\text{m}$
- or Policy ODD departure
- u_{prop} flagged with lease intent

PHASE 2 · INTERCEPT & AUTH

Hardware Verification

- 1 kHz MCU intercepts token
- HMAC-SHA256 & Nonce valid
- Freshness: $\Delta t_{\text{transit}} \leq 100 \text{ ms}$

PHASE 3 · COMMIT & BLEND

Bumpless Ramp

- Atomic swap at 1 ms tick
- C^1 blend: $a(t)$ over τ_{blend}
- Integrators pre-biased to $x(t)$

PHASE 4 · CONFIRM & ACTIVE

Active Authority

- Haptic confirmation feedback
- Human in command ($\alpha = 1.0$)
- Active lease ($T_{\text{lease}} \leq 100 \text{ ms}$)

DISCRETE AUTHORITY STATE MACHINE ($S_{\text{t}} \in \{\text{Auto, Pending, Blend, Manual, Fallback}\}$)

S_{auto} · AUTONOMOUS

Authority: Policy ($\alpha = 0.0$)

- High-level action chunks
- 1 kHz Enforcer bounds
- Background state tracking

S_{pending} · PENDING

Authority: Frozen / Check

- Handshake timer armed
- Token & nonce validation
- Guard: $t \leq T_{\text{timeout}} (500\text{ms})$

S_{blend} · BUMPLESS RAMP

Authority: Dynamic Blend

- C^1 cubic schedule $a(t)$
- Window $\tau_{\text{blend}} \geq 1.5|\Delta u|/j_{\text{max}}$
- Jerk bounded: $j \leq j_{\text{allowable}}$

S_{manual} · ACTIVE HUMAN MANUAL

Authority: Human Operator ($\alpha = 1.0$)

- Direct steering / torque dispatch
- Continuous deadman lease ($T_{\text{lease}} \leq 100 \text{ ms}$)
- Subordinate to Tier-1 Safety Barrier $h(x) \geq 0$
- Shadow logging of background policy inferences

S_{fallback} · DEGRADED FALLBACK

Category 1 Controlled Deceleration

- Deterministic braking: $a_{\text{brake}} = \mu g (6.87 \text{ m/s}^2)$
- Monotonic lock: No upward transitions during motion
- Forensic buffer freeze: 10s pre/post trigger window
- Exit: Zero velocity ($v = 0$) + Signed Re-init

Override Trigger

Auth Verified

Ramp Complete ($\alpha = 1.0$)

Timeout $t > T_{\text{timeout}} (500 \text{ ms})$

Lease Expired / Touch Loss $> 100 \text{ ms}$

$v = 0$ & Reset

3-TIER ARBITRATION & EVIDENCE AUDIT

TIER 1 · ABSOLUTE VETO (MCU REFLEX)

1000 Hz Hard Real-Time Enforcer

CBF Invariant: $h(x) \geq 0$ · Over-current trip

TIER 2 · SUPERVISORY OVERRIDE

Authenticated Human Input ($\alpha = 1.0$)

Bounded deadman lease ($T_{\text{lease}} \leq 100 \text{ ms}$)

TIER 3 · PROPOSAL STREAM (STOCHASTIC)

20–50 Hz Host MPU Action Chunks

Candidate action buffer u_{prop} (Diffusion / ACT)

64-BYTE FORENSIC EVIDENCE TUPLE

$(t_{\text{PTP}}, S_{\text{auth}}, x_{\text{state}}, u_{\text{prop}}, u_{\text{enf}}, u_{\text{act}}, \text{ID}_{\text{op}})$

- IEEE 1588 PTP hardware timestamps ($< 100 \text{ ns}$ uncertainty)
- SHA-256 HMAC hash chaining into write-once ring buffer
- Deterministic logging: $< 4 \mu\text{s}$ latency (zero dynamic malloc)